amazon rme

AMAZON BASE BUILDING MAINTENANCE STANDARD

GLOBAL CENTRAL RME STANDARDS & COMPLIANCE

Task Title	ELECTRICAL SYSTEM, UNINTERRUPTABLE POWER SUPPLY - 11
APM Object	ELEC.UPS
APM EC	AMZ.ELEC.UPS
BBM Handbook Title	Uninterruptable Power Supply (UPS)
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure These tasks are based on industry guidance, SME knowledge and HIE learning. Please ensure vendor specific requirements are followed. To suggest improvements please raise a SIM SIM - Link to SIM template

Last Update: (Version)	Jul-23	**Geo Applied:**	Intl APM Instance	**F-Gas**	Not Identified	**Activies may be covered by No-Go restrictions:**	TBC	**Special Certificaion Required:**	TBC	**Delivered By:**	Carried out by trained on site technician certified in the electrical field	**OCB:**	BBM
Evidence for CT:	TBC	**Pricing template Position:**	11	**Related CT Areas:**	CT.ELEC.UPS			**Assets included in this PM:**	BATTERY BACKUP UPS - RACK MOUNTED UPS	**Schedule With:**	-		

Description of System / Technology (According to BBM Handbook)

Uninterruptible power supply is a type of continual power system that provides automated backup electric power to a load when the input power source is removed. Typically this backup power is supplied by batteries or supercapacitors which gives instantaneous transfer. Each model has a different able to support the load for a set duration depending on design. These are typically supporting network equipment(MDF) or critical process equipment. Rack mounted UPS are small single phase units to supply individual device such as IDF cabinet, arc flash relay, MHE control cabinet gateway.

Additional Information

All sites with the above listed Base Building Asset should conduct routine maintenance and inspections according to this standard PM package. The Amazon maintenance standard supports compliance, reduces safety risks,improves asset reliability & efficiency. Some maintenances captures several assets within one PM package -> Ensure that corrective/reactive maintenance for assets where proactive maintenance is contained under another asset is created against the original asset.

Task Index	Component	Tasks	Tasks Nested for frequencies > weekly									
			Daily	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	36 Monthly	48 Monthly	60 Monthly
1	General check	Is the room properly ventilated						PM				
2	General check	Are the components clean - free from debris						PM				
3	General check	Is the air filter & grid of the cooling inlets clean -> Clean or change filter if necessary					SC	PM				
4	General check	Are the cooling fans operating appropriately and doesn`t make any unusual noise						PM				
5	General check	Visual check of the equipment for damage & missing components					SC	PM				
6	General check	Are there any alarms indicated					SC	PM				
7	General check	Did any circuit breaker trip						PM				
8	General check	Are disconnecting devices lockable in the open position (LOTO)						PM				
9	General check	Are the circuit diagrams and relevant technical documentation available and complete						PM				
10	General check	Are there manual entries in the drawings or technical documentation that indicate changes to the installation						PM				
12	Thermographic scan	Perform thermographic scans of the electrical cabinet and the battery system -> Create a WO for over-temperature to determine the root cause immediately						PM				
13	Thermographic scan value	Add a detailed note for all over-temperature measurements -> list the location and temperature -> Upload the thermographic scans for overtemparature						PM				
14	Electrical components	Visual check of cable connections such as pinched insulation in the contact area						PM				
15	Electrical components	Visual check for the protection against direct contact with live parts						PM				
16	Electrical components	Were any obvious changes made (e.g. rewiring, hanging cables)						PM				
17	Electrical components	Are all cables properly routed						PM				
18	Electrical components	Are all components properly labeled or marked						PM				
19	Electrical components	Test all safety relevant switches						PM				
20	Electrical components	Are all components properly attached						PM				
21	Electrical components	Are all metal parts connected to the protection earth (ground)						PM				
22	Electrical components	Is the ground connection to the building ground as designed?						PM				
23	Electrical components	Ensure all cable connections are correctly torqued according to manufacturers specification						PM				
24	Battery & Charger	Measure the voltage of each battery (disconnected) and plan for replacement if necessary						PM				
25	Battery & Charger	Measure the charging voltage and current before & after functional test						PM				
26	Battery & Charger	Compare to design specification & plan component exchange in case of an issue						PM				
27	Battery & Charger	Check the internal uninterruptable voltage supply of the unit						PM				
28	Battery & Charger	Check the batteries for leakage						PM				
29	Battery & Charger	Check the fluid level of the batteries and recharge if required						PM				
30	Functional test	Simulate a power interruption and test the UPS for correct function (avoid impact on connected assets)						PM				
31	Functional cleaning	Carry out a functional cleaning if required						PM				
33	Final Check	Check the O&M manual and carry out additional required tasks						PM				

Change Log

July 2023 - First issue in new format. Simplification of task scope, alignment with CT.
July 2024 - Task 13 updated.